



A COMPARISON OF ALGORITHMS USED TO COMPUTE HILL SLOPE AS A PROPERTY OF THE DEM

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Abstract—The calculation of hill slope in the form of downhill gradient and aspect for a point in a digital elevation model (DEM), is a popular procedure in the hydrological, environmental and remote sensing. The most commonly used slope calculation algorithms employed on DEM topography data make use of a three by three search window, or kernel, centred on the grid point (grid cell) in question in order to calculate the gradient and aspect at that point. A comparison of eight frequently used slope calculation algorithms for digital elevation matrices has been carried out using both synthetic and real data as test surfaces. Morrison's surface III, a trigonometrically defined surface, was used as the synthetic test surface. This was differentiated analytically to give true gradient and aspect values against which to compare the results of the tested algorithms. The results of the best-performing slope algorithm on Morrison's surface were then used as the reference against which to compare the other tested algorithms on a real DEM. For both of the test surfaces residual gradient and aspect grids were calculated by subtracting the gradient and aspect grids produced by the algorithms on test from the true/reference gradient and aspect grids. The resulting residual gradient and aspect grids were used to calculate root-mean-square (RMS) residual error estimates that were used to rank the slope algorithms from "best" (lowest value of RMS residual error) to "worst" (largest value of RMS residual error). For Morrison's test surface, Fleming and Hoffer's method gave the "best" results for both gradient and aspect. Horn's method (used in ArcInfo GRID) also performed well for both gradient and aspect estimation. However, the popular maximum downward gradient method (MDG) performed poorly, coming last in the rankings. A similar pattern was seen in the gradient and aspect rankings derived using the Rhum DEM, with Horn's method performing well and the MDG method poorly. © 1998 Elsevier Science Ltd. All rights reserved

Key Words: Slope, Gradient, Aspect, Altitude, Elevation, Slope line, Digital elevation model, DEM, Complex numbers.

INTRODUCTION

The use of digital elevation models (DEMs) as a basis for deriving secondary products such as gradient or aspect, exposure or water catchments and intervisibility has been well documented by several authors (Moore and others, 1991; Maidment, 1993; Fisher, 1994). Given the importance of gradient and aspect estimates in many environmental models there is a need rigorously to review the techniques available for calculating these parameters from the initial DEM, taking into account the scale and resolution of the data and the nature of the slope and aspect values that may be derived. In this paper a comparison is made between eight algorithms used to compute gradient and aspect values as a local property of the square lattice/grid type DEM. A DEM is a regular, square array of elevations typi-

cally derived by photogrammetric or cartographic methods (Moore and others, 1991).

DEFINITIONS AND NOMENCLATURE

In order to avoid ambiguity the following nomenclature is used. These definitions have been taken with slight modifications, from Evans (1979, pp. 28–29).

Altitude, elevation: Height above mean sea level, measured in metres (m) or arbitrary units.

Aspect: The horizontal direction of movement down a slope line. The compass direction (measured in degrees from a reference) in which the plane tangent to the surface at a point faces, measured outward from the surface at right angles to the contours.

Gradient: The rate of change of altitude along a slope line (i.e. the change in altitude divided by the change in horizontal position).

Slope: Properties of a plane tangent to a point on a surface. These can be specified in terms of a single normal vector, or both gradient and aspect together.

Slope line: A line of locally greatest rate of change of altitude, along which a frictionless ball starting from rest would roll.

METHODOLOGY

This study first used the 49-term trigonometric surface of Morrison (1971, 1974), (Fig. 1) as the synthetic terrain surface with which to compare the various slope derivation methods. Hodgson (1995) used this surface in a similar way for a study of some slope algorithms. Other investigations of such algorithms (Skidmore, 1989; Onorati and others, 1992) have tended to use values of gradient and aspect manually calculated from paper maps as the “true” values against which to compare slope algorithm results (Skidmore, 1989).

Morrison’s test surface is the summation of the zero through third harmonic in both the x and y base directions and the area of definition is $0 \leq x \leq 1000$ and $0 \leq y \leq 1000$, where x and y are measured in radians. The 49-term equation for the surface represents a least-squares fit to 121 data points read from a square lattice on Hsu and Robinson (1970) surface III, which was a representation of a real topographic surface. Morrison’s surface has a maximum downhill gradient of 0.89 (41.7°) and a mean value of downhill gradient of

0.34 (18.8°). The surface was differentiated analytically in order to obtain the partial derivatives in the x and y directions, respectively. This enables the true values of gradient (the gradient in the down dip direction) and aspect (the azimuth of the down dip direction) at any point on the test surface to be determined from the partial derivatives in the manner described by Sharpnack and Akin (1969). True analytically derived gradient and aspect values for Morrison’s surface were used as a reference against which to compare slope results obtained from the numerical algorithms under test.

Most previous studies of slope algorithms that have been made using Morrison’s surface, (Hodgson, 1995, for example) have used a very fine grid (small cell size) and a numerical method in order to find the reference slope values. The significant advance in the methodology here is the application of analytic differentiation to obtain the true gradient and aspect values.

In this study of slope algorithms using Morrison’s surface, ten cell sizes of between 1/100 and 1/10 (1/100, 1/90, 1/80 etc. to 1/10) of the x and y dimensions of the surface were used to represent different sampling intensities, with DEMs being created to represent Morrison’s test surface at each sampling intensity, starting with a grid size of 101 by 101 and decreasing the grid size by ten for each new test DEM until the smallest DEM/grid size of 11 by 11 elevation points was reached. In a similar manner grids of analytic gradient values were created in order to represent the true gradient data values. In order to eliminate any directional bias in the test data, Morrison’s surface was rotated through steps of twenty degrees in azimuth to pro-

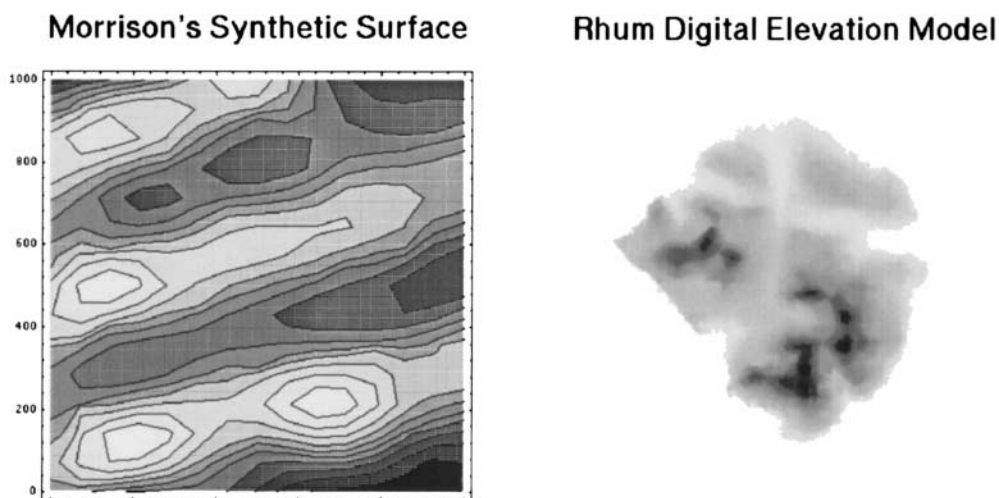


Figure 1. Morrison’s synthetic surface and the Rhum digital elevation model. The synthetic surface was generated using a 49-term trigonometric series after Morrison (1971, 1974), with the darkest shading being used for the lowest elevations. The digital elevation model shows a 100 m grid point spacing elevation matrix representing the Isle of Rhum, in the form of a grid of 141 by 151 elevation points. The Rhum elevation model is represented here with the darkest shading used for the highest elevations.

duce eighteen ensembles of the required sampling intensities. The coordinate transform required to rotate the surface was based on that described by Pavlidis (1982). Due to time constraints, the study of aspect results from the eight methods on test was carried out using Morrison's surface represented by only a single 101 by 101 point DEM grid. A corresponding grid of analytically derived aspect values was created in order to represent the true aspect data values.

The eight slope algorithms under scrutiny were applied to each of the test DEMs corresponding to each of the different sampling intensities and orientations considered. The resulting grids of numerically derived gradient values were each subtracted from those of the corresponding analytic gradients in order to produce outputs of differences between true and numerical gradient values, which have been termed "residual gradient grids". The root-mean-square (RMS) average value of each residual gradient grid was calculated in order to give an estimate of the error in the numerical gradient values for each slope algorithm, at each cell size. In a similar manner the aspect results of the eight numerical algorithms tested using the 101 by 101 point representation of Morrison's surface were compared with the corresponding analytic aspect grid, in order to form eight residual aspect moduli grids and corresponding RMS error values. The RMS error values generated for the grids of both gradients and aspects, were used to rank the slope algorithms from "best" to "worst". Finally, a relative comparison of slope algorithms was made using a DEM of a real topographic surface. A comparison was made between the "best" performing algorithm on Morrison's surface and the other slope algorithms tested. Using a real DEM in this way enabled the evaluation of the validity of the ranking of slope algorithms using Morrison's surface. The real dataset we used was a DEM of the Island of Rhum in the Inner Hebrides, which took the form of a grid of 141 by 151 elevations, the grid points being spaced at 100 m intervals. Figure 1 shows a representation of this surface.

NUMERICAL SLOPE ALGORITHMS

The principal difference between most algorithms that compute slope as a local property of the DEM is the number of grid cell values used and the weighting given to each of these cell values. The most frequently used algorithms employ between four and nine of the elevation values in a three by three window, or kernel, centred on the elevation cell in question in order to calculate a estimates of gradient and aspect.

The eight slope algorithms evaluated were:

(a) The method of Fleming and Hoffer (1979), which was further described by Unwin (1981) and

presented in the form of an algorithm by Ritter (1987). This is the so called "rook's case", a widely used second-order, finite-difference algorithm that computes gradient and aspect from only the nearest four elevation points on the grid. This method calculates an east-west gradient for a given elevation cell from the "rise over run" between the adjacent cells on either side, in that direction. The north-south gradient is obtained in a similar way. The values of gradient and aspect can be calculated from these perpendicular apparent gradients in a simple manner (Sharpnack and Akin, 1969). This method uses four grid points to calculate each gradient and aspect value. Zevenbergen and Thorne (1987) provide a novel derivation of this method in their deducing of the coefficients of a partial quadratic trend surface which passes exactly through the nine elevation points of the three by three DEM sampling kernel.

(b) The method of Horn (1981) is a modified version of the technique of Sharpnack and Akin (1969), using unequal weighting coefficients for the nearer elevation values. These weightings are proportional to the reciprocal of the square of the distance from the kernel centre. Horn's method is implemented in the "SLOPE" function of the ArcInfo GRID software package (ESRI, 1995). This method uses eight grid points to calculate each gradient and aspect value.

(c) The "one over distance" weighting method is similar to that of Horn (1981), but differs in that the weighting coefficients used are proportional to the reciprocal of the distance from the kernel centre as described by Unwin (1981), rather than the reciprocal of the square of the distance. This method also uses eight grid points to calculate each gradient and aspect value.

(d) The algorithm of Sharpnack and Akin (1969) employs a third-order finite-difference technique using the eight neighbouring elevation values bordering the central elevation cell. This method uses eight grid points to calculate each gradient and aspect value. The gradient and aspect values produced are the same as those for a least squares linear trend surface fitted to the eight elevation values (i.e. linear regression) (Sharpnack and Akin, 1969; Wood, 1996, ch. 4). In addition, it has been shown by Evans (1979) that terrain first derivatives (gradient and aspect) derived from an unconstrained quadratic surface fitted by least squares to the elevation values in the three by three DEM sampling kernel are also the same as those given by Sharpnack and Akin's method.

(e) The constrained quadratic surface method (Wood, 1996, ch. 4). This algorithm uses a quadratic regression surface that is constrained to go through the central elevation point of the local three by three DEM sampling kernel. The method creates a local X - Y - Z origin in the middle of the central cell. This causes all other elevations in the

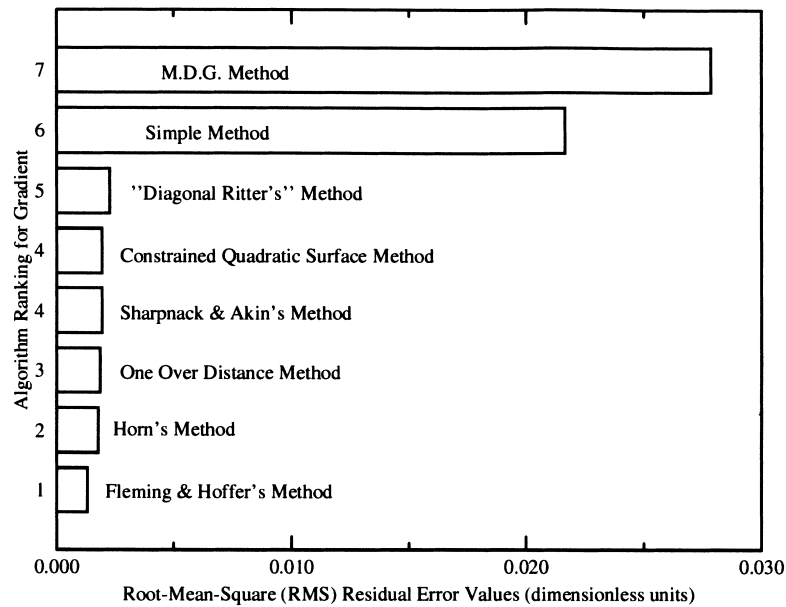


Figure 2. RMS residuals between ideal and calculated gradients, on Morrison's unrotated surface, for a gridsize of 101 by 101 points.

sampling kernel to be expressed as relative vertical differences from the central value. The fitted trend surface is forced to go through this origin. The partial first derivatives of the fitted quadratic trend surface at the origin, provide values of apparent gradients for both the X and Y directions from which aspect and downhill gradient may be calculated. The second derivatives of the fitted quadratic surface can be used to calculate curvature values, such as plan and profile curvatures.

(f) The "diagonal Ritter's" method is a modification of that of Fleming and Hoffer (1979). The novel feature is that the perpendicular apparent gradients are taken at an angle of 45° to the principle axis of the DEM grid. This results in a square root of two multiplication in the effective "ruler-length" of the method. This method uses

four grid points to calculate each gradient and aspect value.

(g) A method termed here as the simple method, is similar to that of Fleming and Hoffer (1979), but uses the difference in elevation between a given elevation cell and its neighbour to the west to calculate the "east-west gradient" and a similar technique to get the "north-south gradient". This method uses three grid points to calculate each gradient and aspect value.

(h) The maximum downward gradient method (Travis and others, 1975; EPPL, 1987; Onorati and Poscolieri, 1988). This technique selects the largest downward gradient angle from those computed by comparing the elevations of every grid cell in the DEM with those of the eight nearest neighbours. The aspect value given by this method can have one of eight distinct values, i.e. 0, 45, 90, 135, 180, 225,

Table 1. Slope algorithm rankings derived using Morrison's surface. Ranking is from 1 ("best") to 7 ("worst")

Ranking	Residuals used to generate RMS residual error values in order to obtain the method rankings	
	residual gradient grids	residual aspect moduli grids
1	Fleming and Hoffer's	Fleming and Hoffer's
2	Horn's	Horn's
3	one over distance weighting	one over distance weighting
4	Sharpnack and Akin's	Sharpnack and Akin's
4	constrained quadratic	constrained quadratic
5	"diagonal Ritter's"	"diagonal Ritter's"
6	simple method	simple method
7	maximum downward gradient	maximum downward gradient

270 and 315° corresponding to the azimuth of the eight nearest neighbours from the central elevation cell in the kernel. The algorithm represents a simple and fast method for analysing a three by three cell window (kernel). The method uses elevation information from all nine grid points in order to select two from which to calculate each gradient and aspect value.

RESULTS USING MORRISON'S SURFACE

Gradient

The results of this investigation using the unrotated Morrison's surface suggest that the various slope algorithms tested can be ranked in terms of RMS residual error for gradient, from lowest ("best") to highest ("worst") effective error values (Table 1). This ranking is clearly seen (Fig. 3) in the results for grid sizes of between 21 by 21 and 101 by 101 points used to represent the area of definition of the test surface. Figure 2 shows the algorithm ranking for gradient, in terms of RMS residual error for the 101 by 101 point grid, and Figure 3 shows how the RMS residual error values vary with grid size between 101 by 101 and 11 by 11 points.

The investigation also included slope algorithm rankings calculated using Morrison's surface rotated through steps of 20° in azimuth to produce eighteen ensembles of test DEMs of the required sampling intensities. Between grid sizes of 101 by 101 and 41 by 41 elevation points, the algorithm rankings for gradient for all azimuths are the same as those given for the unrotated case previously discussed. Figure 4 shows a graph of RMS residual error as a function of clockwise rotation angle (azimuth) for a grid size of 101 by 101.

Between grid sizes of 41 by 41 and 21 by 21 a similar pattern is seen to that outlined in the previous paragraph, but with the additional feature that for certain discrete rotation angles, the simple

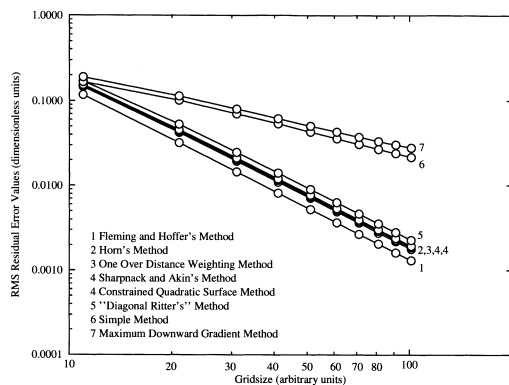


Figure 3. RMS residuals between ideal and calculated gradients, on Morrison's unrotated surface, for gridsize of between 11 by 11 and 101 by 101 points.

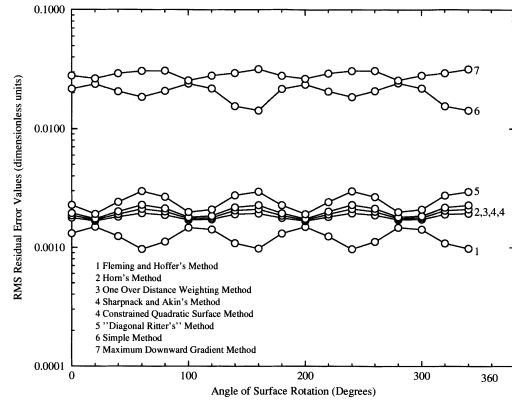


Figure 4. RMS residuals between ideal and calculated gradients, versus angle of rotation of Morrison's surface, for gridsize of 101 by 101 points.

method will slightly overlap with either the "diagonal Ritter's" or maximum downward gradient method. This can be seen in Figure 5, which shows RMS residual errors versus azimuth for a grid size of 21 by 21 points. For grid sizes smaller than 21 by 21 points the situation is more complex, and when a grid size of 11 by 11 points is reached most of the RMS residual error versus azimuth curves given by the various slope algorithms are overlapping. For this small grid size no meaningful slope algorithm ranking can be discerned apart from the observation that Fleming and Hoffer's is the "best" method and the maximum downward gradient method is the "worst".

Aspect

In order to calculate the residual aspect grids a two dimensional vector representation was used for each aspect value, taking the form of a unit length vector contained in the X - Y plane, oriented in the given aspect direction (after Agterberg, 1974,

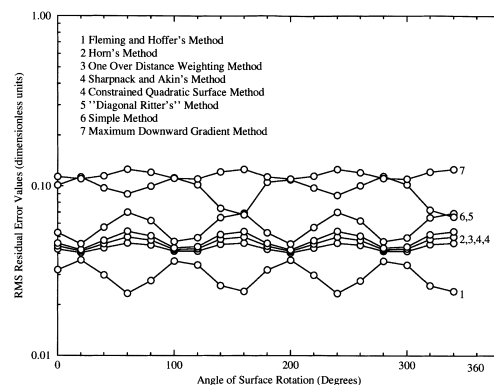


Figure 5. RMS residuals between ideal and calculated gradients, versus angle of rotation of Morrison's surface, for gridsize of 21 by 21 points.

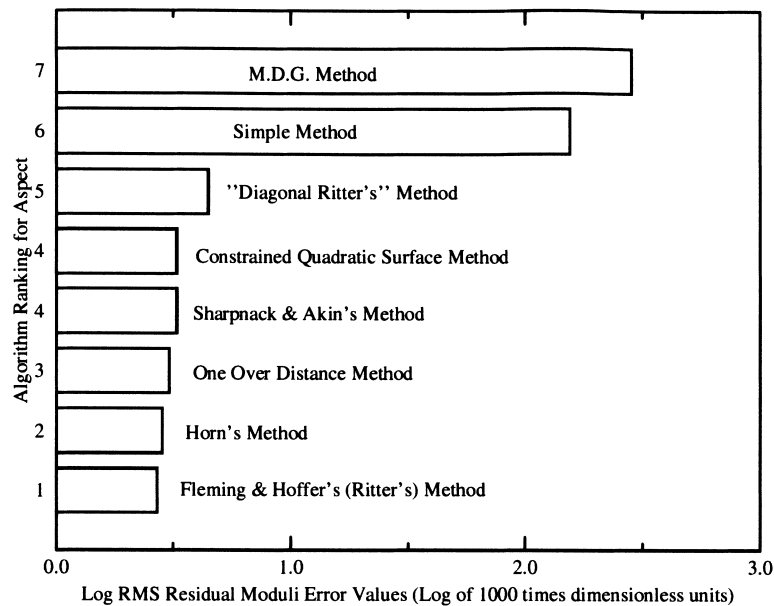


Figure 6. RMS residuals between ideal and calculated aspect vectors, on Morrison's unrotated surface, for a gridsize of 101 by 101 points. RMS residuals are calculated from moduli of residual aspect vectors.

pp. 475–508). Since each aspect vector could be described by its perpendicular components in the X and Y directions, a representation was chosen that expressed each vector as a single complex number, the perpendicular components of the vector being represented by the real and imaginary parts. The

RMS residual error for aspect was calculated as follows. First, a residual aspect vector grid was formed by subtracting a grid of numerically calculated aspect vectors from the reference analytically calculated aspect vector grid. Second, a grid of vector magnitude (moduli) values was calculated from the

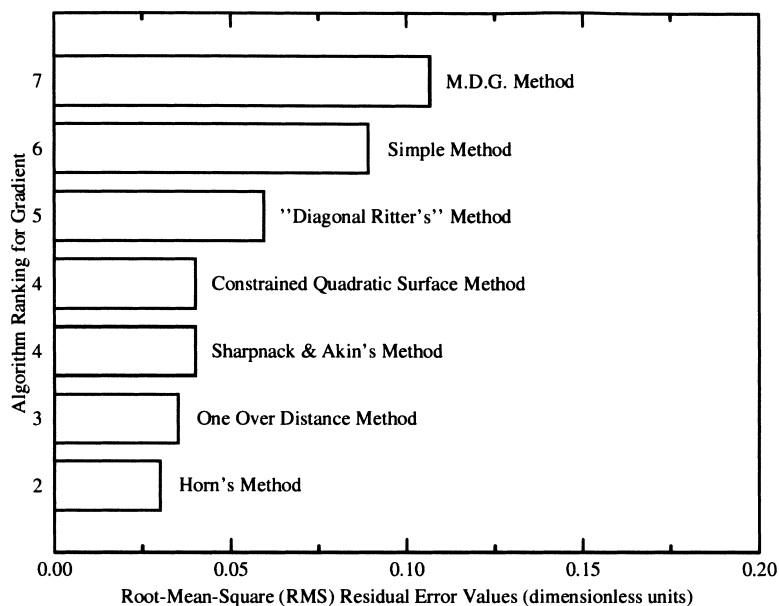


Figure 7. RMS residuals between gradients calculated using Fleming and Hoffer's method and gradients calculated using other methods tested, for digital elevation model (DEM) of Isle of Rhum made up of 141 by 151 elevation points spaced at 100 m intervals.

vectors in the residual aspect vector grid. This grid of moduli values, which are scalar numbers, was used to calculate a single RMS error estimator, termed the RMS residual moduli error value. The RMS residual moduli errors for aspect give a ranking for the tested slope methods as shown in Figure 6. The algorithm ranking obtained for aspect grids using Morrison's surface is the same as that for gradient (see Table 1).

RESULTS USING A 100 M INTERVAL DEM FOR RHUM

Using a DEM of the Isle of Rhum made up of a grid of 141 by 151 elevation points spaced at 100 m intervals (Fig. 1), a comparison was made between the results of Fleming and Hoffer's method and the other seven slope algorithms tested. Fleming and Hoffer's method was used as the reference method because it was the "best" performing algorithm for both gradient and aspect in terms of RMS residuals on Morrison's test surface. Using a DEM of a test area in this way enabled us to evaluate the validity of ranking of slope and aspect algorithms using Morrison's surface. Figure 7 shows a histogram of the RMS residual error values derived from the gradient grids output by the various slope algorithms tested on the Rhum DEM, plotted in the order of the original gradient ranking using Morrison's surface. The Rhum DEM was also used to rank the slope algorithms for aspect. Residual aspect grids were created by subtracting the aspect vector results of a given algorithm from those given by Fleming

and Hoffer's method. The resulting RMS residual moduli aspect error value being calculated for each grid (Fig. 8).

The rankings obtained for both gradient and aspect using the Rhum DEM are the same as those obtained for both gradient and aspect on Morrison's surface (see Figs 2 and 6–8 and Table 1). Fleming and Hoffer's method has been defined *a priori* as being the best method in the Rhum DEM case, and hence is not included in the resulting rankings. The agreement of the order of slope algorithm rankings on both surfaces thus gives enhanced credibility to slope ranking results obtained from Morrison's surface.

The maximum gradient value derived from the Rhum DEM lies in the range 1.48 (56.0°) for Sharpnack and Akin's method, to 3.04 (71.8°) for the simple method, depending on the slope algorithm used. The mean value of gradient for the Rhum DEM was approximately 0.12 (6.8°) irrespective of the slope algorithm used.

DISCUSSION

The slope algorithm comparisons on both Morrison's synthetic surface and the Rhum DEM show that within the constraints of the numerical techniques used, the constrained quadratic surface method and Sharpnack and Akin's method give identical results for gradient and aspect estimates for a given input and should thus be considered as equivalent methods. Although these two methods may be degenerate as far as first derivatives of the

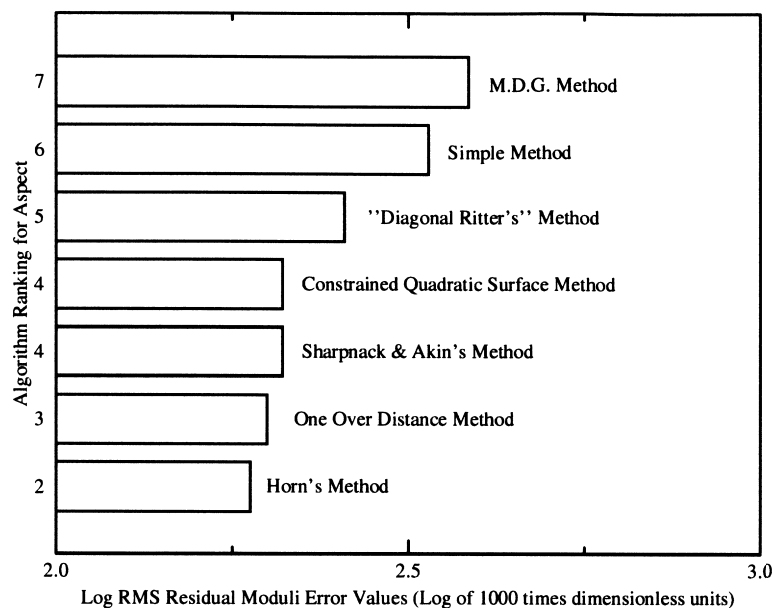


Figure 8. RMS residuals between aspect vectors calculated using Fleming and Hoffer's method and aspect vectors calculated using the other methods tested, for digital elevation model (DEM) of Isle of Rhum made up of 141 by 151 elevation points spaced at 100 m intervals. RMS residuals are calculated from moduli of residual aspect vectors.

terrain surface is concerned, it should be remembered that of the two, only the constrained quadratic surface method is capable of providing second derivatives of terrain, such as profile and plan curvatures. The author was unable to find an analytic proof in the literature that the constrained quadratic surface method is equivalent to Sharpnack and Akin's method for first derivatives of terrain. However, an analytic proof by Evans (1979) exists showing that the partial first derivatives of terrain derived from an unconstrained quadratic surface fitted to a local three by three DEM sampling kernel by least squares are the same as those obtained from Sharpnack and Akin's method. Further, an analytic proof also exists (Sharpnack and Akin, 1969) demonstrating that an unconstrained linear trend surface fitted by least squares to the local three by three sampling kernel (i.e. linear regression) also gives the same results for terrain partial first derivatives as Sharpnack and Akin's method. These proofs together with the numerical results from this study strongly suggest that any quadratic surface whether constrained to agree with the central elevation point of the kernel or not fitted by least-squares to the local three by three DEM sampling window will yield the same gradient and aspect results as Sharpnack and Akin's method.

The ranking of slope algorithms from "best" to "worst" derived using Morrison (1971, 1974) test surface do not contradict the results of Hodgson (1995). He studied the same surface, as far as methods using either four or eight kernel elevation values for each slope estimate were concerned. Hodgson used a comparison of bi-directional surface normal vectors (which contain both gradient and aspect information) to evaluate several slope algorithms. He concluded that algorithms employing only four neighbouring cell values were consistently more accurate at estimating surface normal vector directions (and hence gradient and aspect values) than algorithms using eight neighbouring cell values. This study of gradient and aspect values derived from various numerical slope algorithms applied to Morrison's surface, concurs with Hodgson's findings, with the "best" algorithm considered (Fleming and Hoffer's) using four neighbouring cell values, and the next three "best" algorithms (Horn's, one over distance weighting method and Sharpnack and Akin's) all employing eight neighbouring cell values (Table 1). A similar dichotomy exists in the slope rankings derived from the Rhum digital elevation model. The "diagonal Ritter's" method does not fit into this pattern. This was as expected since taking the perpendicular partial gradients at an angle of 45° to the principle axis of the DEM grid will cause a square root of two multiplication in the effective "ruler-length" of the method. This explains why despite using just four neighbouring cell values to calculate each slope estimate, the values of RMS average residual associ-

ated with this method are of a magnitude more typical of eight neighbour slope methods.

The results in this paper also show that methods that employ only one or two nearest neighbours and the central value in general perform less well than methods using eight or four neighbouring elevation points. These findings are valid for both Morrison's surface and the Rhum DEM, and thus it is suspected that they are generally applicable to terrain surfaces. The maximum downward gradient method, which effectively uses two elevation points to calculate slope, gave the "worst" results for of any algorithm tested on Morrison's surface or the Rhum DEM.

We conclude that in general there does not seem to be a simple relationship between the number of grid points used to calculate a gradient or aspect value by an algorithm and the root-mean-square residuals associated with that algorithm. However, it can be seen from Figure 3 that the slope algorithms fall into two clear groupings with the simple method and the maximum downward gradient method having similarly high RMS residual values compared with all the other methods tested for a wide range of DEM resolutions. Figures 4 and 5 demonstrate the relative insensitivity of the methods to the different orientations of Morrison's test surface.

The justification for using Fleming and Hoffer's method as the standard or reference method for use with the Rhum DEM is that it performs best on Morrison's surface. This justification is only valid if the assumption is made that both Morrison's surface and the Rhum DEM have high values of spatial autocorrelation. However, since high autocorrelation values are characteristic of real terrain surfaces at the scales they are usually represented on topographic maps and associated DEMs, this assumption is not unreasonable, especially when one remembers that Morrison's synthetic surface is itself a representation of the real topographic surface of Hsu and Robinson (1970). However, for surfaces with very much higher levels of surface roughness, and consequently lower values of spatial autocorrelation, the author concedes that this assumption may break down and a different technique of slope algorithm ranking be required.

A feature of slope algorithm investigation that has not yet been addressed is the comparison of algorithm performance in various gradient classes from almost flat slopes with small gradients, up to almost vertical slopes with large gradient values. This study, by contrast, has used the entire extent of the provided test surfaces in order to create slope algorithm rankings. This is valid for choosing a slope algorithm for use on a DEM representing topography with a wide range of gradient values.

Alternative methods may be used to continue the analysis of slope algorithms from the study of various synthetic and real terrain surfaces. For

example, Gaussian noise could be used to simulate a rough or noisy terrain surface on which slope algorithms could be compared. This approach could be extended by adding various amounts of Gaussian noise to an already existing smooth surface, such as Morrison's surface III, in order to give slope algorithm rankings appropriate to the whole spectrum of surface roughnesses. In addition to the scale related work done in this study using DEMs of different grid sizes to represent Morrison's synthetic surface, studies of the relative performance of slope algorithms on DEMs derived from a wide range of scales of real topographic mapping should be considered, in order to address effects that scale changes in real data might have on the slope algorithm ranking.

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